COHOMOLOGY OF A SITE EQUALS COHOMOLOGY OF A BIGGER SITE UNDER THE COMPARISON HYPOTHESES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

Editorial note. Both referenced results live in earlier chapters, so their locators point at the verbatim lines of the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt) that carry the canonical tags.

Allowed standard prerequisites (editorial).

- The comparison hypothesis on a functor of sites and the induced morphism of topoi. Sites, Lemma labelled sites-lemma-bigger-site (canonical Stacks tag 00XU); named in the statement at line 650 and again in the proof at line 677. Supplies what the hypothesis on u says, in particular the equality of global sections in degree zero and the identification of the left adjoint of g^{-1} with g_! = f^{-1}.
- An adjoint pair with exact left adjoint preserves injectives. Homology, Lemma labelled homology-lemma-adjoint-preserve-injectives (canonical Stacks tag 015Z); invoked at proof lines 673-674. Supplies that g^{-1} carries injective abelian sheaves to injective abelian sheaves; the exactness check on the left adjoint is retained in the counted span.

Source reference. Original lemma 00XU: sources/stacks-chow/tags.txt, source lines 1232–1232. Cross-chapter pointer named in the counted statement at line 650 and used again at proof line 677; it supplies the comparison hypotheses and the notation for the left adjoint, while the injective-preservation argument is retained in the counted span.

Source reference. Original lemma 015Z: sources/stacks-chow/tags.txt, source lines 1526–1526. Cross-chapter standard prerequisite invoked once at proof lines 673-674 for the criterion that an adjoint pair with exact left adjoint preserves injectives; the exactness check on g_! is retained in the counted span.

Lemma 0.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be sites. Let $u : \mathcal{C} \to \mathcal{D}$ be a functor. Assume u satisfies the hypotheses of Sites, Lemma 00XU. Let $g : Sh(\mathcal{C}) \to Sh(\mathcal{D})$ be the associated morphism of topoi. For any abelian sheaf $\mathcal{F}$ on $\mathcal{D}$ we have isomorphisms*

$$R\Gamma(\mathcal{C}, g^{-1}\mathcal{F}) = R\Gamma(\mathcal{D}, \mathcal{F}),$$

in particular $H^p(\mathcal{C}, g^{-1}\mathcal{F}) = H^p(\mathcal{D}, \mathcal{F})$ and for any $U \in \mathrm{Ob}(\mathcal{C})$ we have isomorphisms

$$R\Gamma(U, g^{-1}\mathcal{F}) = R\Gamma(u(U), \mathcal{F}),$$

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in particular $H^p(U, g^{-1}\mathcal{F}) = H^p(u(U), \mathcal{F})$. *All of these isomorphisms are functorial in* $\mathcal{F}$.

Proof. Since it is clear that $\Gamma(\mathcal{C}, g^{-1}\mathcal{F}) = \Gamma(\mathcal{D}, \mathcal{F})$ by hypothesis (e), it suffices to show that g^{-1} transforms injective abelian sheaves into injective abelian sheaves. As usual we use Homology, Lemma 015Z to see this. The left adjoint to g^{-1} is $g_! = f^{-1}$ with the notation of Sites, Lemma 00XU which is an exact functor. Hence the lemma does indeed apply. □